ASSIGNMENT 3 - TASK 6

DISK MANAGEMENT FOR A HOSPITAL RECORDS SERVER

Q6 - Disk Management | 4 Marks | Bloom's Level: K5 - Evaluate

1. Problem Context and Design Objectives

The assignment describes a hospital records server that must start reliably, survive disk failure and provide enough swap space for peak load. Storage design must therefore balance data availability, usable capacity, read/write performance, boot reliability and recovery. This report follows Q6 - Disk Management on pages 29-30 of the supplied Assignment 3 question paper and includes the RAID 6 alternative.

2. Disk Preparation: Low-Level, Partitioning and Logical Formatting

Stage	Meaning	Purpose / example
Low-level formatting (physical formatting)	Organises the disk surface into addressable sectors/blocks and establishes low-level control information. On modern drives this is generally performed by the manufacturer; operating systems normally do not perform user-level low-level formatting.	Makes storage addressable by the controller. Defective sectors may be remapped by drive firmware.
Partitioning	Divides a physical disk into logical regions and records boundaries in a partition table such as GPT or MBR.	Creates areas for an EFI System Partition, OS/data volumes, recovery or other purposes.
Logical formatting (file-system creation)	Creates file-system structures inside a partition or logical volume, such as allocation metadata, directories and free-space tracking.	Examples include ext4, XFS or NTFS. Files and directories can then be stored and managed.

These are different operations. Partitioning does not itself create a file system, and creating a file system does not physically format the disk. In a protected server, partition and file-system operations must be planned and backed up because the wrong target device can destroy data.

3. Boot Block and the Boot Process

The boot block is a disk area or boot-related structure containing the initial code or data required to start the boot loader. In a traditional BIOS/MBR system, firmware reads boot code from the boot device and executes it. In a modern UEFI/GPT system, firmware loads an EFI executable from the EFI System Partition (ESP); the boot manager selects a boot entry and starts a boot loader.

Order	Boot stage
1	Power-on and hardware initialisation: BIOS/UEFI performs POST and discovers boot devices.
2	Firmware boot selection: BIOS executes boot code, or UEFI loads an EFI application from the ESP.
3	Boot loader: loads the operating-system kernel and initial RAM file system (initramfs/initrd) when used.
4	Kernel initialisation: detects hardware, initialises drivers and memory management, and mounts the root file system.
5	Init/system manager: starts services, networking, storage services and the hospital records application.

For reliable startup, use redundant boot-capable storage where supported, keep a tested recovery path, monitor boot-device health and maintain verified backups. RAID is not a substitute for backup

or a tested boot recovery procedure.

4. Bad-Block Handling

A bad block is a sector or storage block that cannot be read or written reliably. Modern drives detect defects through internal error-correction and health mechanisms. Depending on the device, firmware can remap a failing logical block to a spare physical sector. The operating system and file system can also detect I/O errors, mark unusable blocks or allocation units, and avoid allocating them again. For SSDs, the flash translation layer manages physical flash blocks, wear levelling and retirement of failing blocks.

A server should monitor SMART/NVMe health and storage error logs, replace drives showing predictive failure, run file-system or array consistency checks as appropriate, and restore damaged data from a verified backup. Bad-block remapping does not recover data that has already become unreadable.

5. Swap-Space Size and Location

Swap provides backing storage for memory pages that are not currently resident in RAM. It can absorb temporary memory pressure, but it is much slower than RAM and should not be treated as a performance substitute for sufficient physical memory. The assignment does not specify server RAM, peak committed-memory demand or hibernation requirements, so an exact mathematical swap size cannot be derived from the source alone.

Proposed starting point: provision an 8 GB swap file on the redundant SSD-backed RAID 10 volume, then validate it under peak-load tests. Final sizing should be based on measured peak commitment:

Required swap ~= max(0, peak committed memory - RAM available to workloads) + safety margin.

Increase the initial 8 GB only if workload measurements show a larger peak deficit. Configure memory-pressure alerts and avoid paging critical database working sets during normal operation. The swap area should be on storage protected against a single disk failure, and the server must continue to boot if a swap device is temporarily unavailable.

Option	Advantages	Disadvantages
Separate physical swap disk	Can isolate paging I/O from the records array; a dedicated fast SSD can improve swap throughput under pressure.	Extra hardware/cost; a single unprotected disk becomes a failure point; swap becomes unavailable if that disk fails.
Swap file on RAID 10 volume	Easy to create/resize; inherits the underlying RAID 10 disk redundancy; no separate disk is required.	Shares I/O with records data; throughput depends on array load and file-system configuration.

6. RAID Comparison

RAID level	Usable capacity (6 x 4 TB)	Fault tolerance	Performance / trade-off
RAID 0	24 TB	No disk failure can be tolerated; failure of any disk can lose the array.	High sequential read/write throughput from striping; no redundancy.
RAID 1 (six-way mirror)	4 TB	The array retains data while at least one of six copies remains.	Reads can be served by multiple mirrors; writes go to all copies. Very strong copy redundancy but low capacity.
RAID 5	20 TB	Any one disk failure can be tolerated; a second failure before recovery can lose data.	Good reads; parity calculation and parity updates add write overhead.
RAID 10 (three mirrored pairs striped)	12 TB	Can survive one disk in each mirror pair; two failures in the same pair lose data.	Strong read/write performance and no parity calculation; less usable capacity than RAID 5/6.

RAID 1 capacity depends on layout. The table treats RAID 1 as one six-way mirror, which provides one disk's capacity (4 TB). Three two-disk mirrors striped together are RAID 10 and provide 12 TB. Capacities are decimal drive capacities before file-system and metadata overhead.

6.1 Recommended RAID level

For the hospital records server, RAID 10 is recommended among the four main options, assuming 12 TB usable capacity meets current and projected records-storage needs. It combines mirroring and striping, provides high read/write performance for database-like workloads, and avoids the parity-update penalty of RAID 5. It can survive one disk failure in each mirror pair, although it cannot survive both members of the same pair failing. Maintain independent, tested backups and replace failed drives promptly.

If the hospital needs more capacity and wants to survive any two drive failures, RAID 6 is a strong alternative. If that protection is more important than write latency, RAID 6 may be preferred; see Section 11.

7. Capacity Calculations for Six 4-TB Disks

Configuration	Calculation	Usable capacity
RAID 0	6 x 4 TB	24 TB
RAID 1 - one six-way mirror	1 x 4 TB	4 TB
RAID 5	(6 - 1) x 4 TB	20 TB
RAID 10 - three mirrored pairs	(6 / 2) x 4 TB	12 TB
RAID 6	(6 - 2) x 4 TB	16 TB

The calculation assumes all disks are the same size and uses the smallest disk size in each group. Actual formatted capacity is lower after file-system metadata and other overhead. RAID 10 tolerance depends on which disks fail: one disk in each pair can be lost, but both disks in one mirror pair cannot.

8. Simplified Probability of Data Loss: RAID 1 and RAID 5

The question asks for a disk-failure rate but does not supply a numeric rate or observation interval. To make the requested calculation reproducible, this report uses an illustrative per-disk failure probability p = 0.02 (2%) over one selected assessment interval. It assumes all drive failures are independent, are counted within the same interval, and no failed disk is rebuilt or replaced during that interval. This is a simplified model, not an annual reliability prediction or a substitute for manufacturer field data.

For one six-way RAID 1 mirror, the array loses all data only if all six copies fail:

P(RAID 1 data loss) = p^6 = 0.02^6 = 0.000000000064

For RAID 5, one failed disk is tolerated. Data is lost if two or more of the six disks fail during the same interval:

P(RAID 5 data loss) = 1 - P(0 failures) - P(1 failure)
= 1 - (1 - p)^6 - 6p(1 - p)^5

Quantity	Value with p = 0.02
P(0 disk failures)	(0.98)^6 = 0.885842380864
P(exactly 1 failure)	6 x 0.02 x (0.98)^5 = 0.108470495616
RAID 1 loss (all 6 mirror copies fail)	6.4 x 10^(-11) = 0.0000000064%
RAID 5 loss (2 or more failures)	0.00568712352 = 0.568712352%

This comparison highlights the value of mirroring, but the simplified RAID 1 result assumes a single six-way mirror. For a conventional two-disk RAID 1 mirror, the probability that both copies fail in the interval is p^2 = 0.0004 (0.04%) under the same p = 0.02 assumption. A three-pair striped mirror is RAID 10 and loses data if both disks in any one pair fail; its simplified array-level probability is 1 - (1 - p^2)^3 = 0.001199520064 (0.119952%). Correlated failures, shared power/controller failures, latent sector errors and rebuild exposure can raise real-world risk. Use the calculator with the failure probability specified by the faculty or supported by device reliability data.

9. Linux Partition and Swap Inspection

The question specifically asks to show the partitions and swap of the Linux system. The script run_task6.sh captures the following read-only inspection commands:

```
lsblk -o NAME,SIZE,TYPE,FSTYPE,MOUNTPOINTS,ROTA,MODEL
sudo fdisk -l
swapon --show
cat /proc/swaps
df -hT
```

The generated files are disk_evidence/disk_layout.txt, fdisk_output.txt and filesystem_usage.txt. The actual output must be collected on the student's Ubuntu/WSL or Linux machine. A WSL environment may show virtual block devices and swap managed by the host; report exactly what is displayed and do not present it as a physical server configuration.

10. Read-Speed Measurement: SSD and HDD

If both device types are available, the script accepts a writable mount directory on each device through SSD_DIR and HDD_DIR. It creates a temporary test file (128 MiB by default), flushes it to storage, then reads it using dd with direct I/O where supported. If direct I/O is unsupported, it attempts a buffered read and labels it as such. The temporary test file is removed after the measurement.

Example (replace the placeholders with mount points reported by lsblk):

```
SSD_DIR="/path/on/ssd" HDD_DIR="/path/on/hdd" TEST_SIZE_MB=128 ./run_task6.sh
```

Device	Mount point	I/O mode	Read speed
SSD	Fill from disk_evidence/SSD_read_result.txt	Direct or buffered	Record local result
HDD	Fill from disk_evidence/HDD_read_result.txt	Direct or buffered	Record local result

No SSD or HDD read-speed value is fabricated in this report because the physical devices available on the user's computer are not visible from the assignment PDF. If only one device type is available, state that a direct comparison could not be performed. Compare results only when test size, block size and I/O mode are the same.

11. Alternative Concept: RAID 6 Instead of RAID 5 or RAID 10

RAID 6 stores two independent parity values distributed across the drives. With six 4-TB disks, two disks' capacity is used for parity:

RAID 6 capacity = (6 - 2) x 4 TB = 16 TB.

RAID 6 can survive any two simultaneous disk failures, unlike RAID 5, which tolerates one. Compared with RAID 5, each small write must update two parity values rather than one. This adds parity computation (commonly XOR plus a second independent coding calculation such as Reed-Solomon/Galois-field operations) and extra read/modify/write I/O. Thus RAID 6 generally has higher write latency and CPU/controller work, although reads remain striped and the extra fault tolerance is valuable.

Feature	RAID 5	RAID 6
Parity	One distributed parity value	Two distributed parity values
Capacity (6 x 4 TB)	20 TB	16 TB
Disk failures survived	Any one	Any two
Small-write overhead	One parity update; read/modify/write penalty	Two parity updates and additional parity calculation/I/O
Use case	Capacity-efficient array where one-drive tolerance is sufficient.	Higher fault tolerance where surviving a second failure during rebuild is important.

If the hospital chooses RAID 6 over RAID 10, it gains 4 TB of usable capacity and can survive any two disk failures, but sacrifices some write performance due to dual parity. RAID 10 remains preferable for write-latency-sensitive workloads when 12 TB is sufficient and mirror-pair failure exposure is acceptable. For this report's main recommendation, RAID 10 is selected because the prompt requires reliable records access and the assumed workload benefits from fast writes; RAID 6 is the stronger alternative when any-two-disk-failure tolerance is the priority.

12. Engineering Recommendation and Conclusion

For the stated hospital records server, use six matched enterprise drives in RAID 10, assuming 12 TB usable capacity meets retention and growth needs. RAID 10 provides redundancy and high read/write performance without parity calculation. Place the swap file on the RAID 10-backed SSD file system, starting with 8 GB and sizing it after peak committed-memory measurements. Use system monitoring, predictive disk-health checks, prompt replacement, tested backups and a documented recovery/boot procedure.

If the hospital's primary requirement is surviving any two disk failures rather than maximising write performance, RAID 6 is the recommended alternative: it provides 16 TB and any-two-disk tolerance, at the cost of extra parity computation and write overhead. RAID 0 is unsuitable because it has no redundancy; RAID 5 is less attractive for write-heavy critical records because it tolerates only one disk failure and adds parity-update overhead. RAID 1 has strong mirror redundancy but only 4 TB capacity under the six-way mirror assumption.

The report's capacity values and probability examples are analytical calculations. The final Linux partition/swap inventory and SSD/HDD read-speed comparison must be completed using the generated evidence files from the user's own system.

13. Execution Procedure and Evidence

1. Place raid_calculator.py, run_task6.sh and submission_evidence_template.txt in one Task06 folder.
2. Open Ubuntu/WSL in that folder.
3. Run chmod +x run_task6.sh.
4. Run ./run_task6.sh; enter the sudo password if prompted for fdisk -l.
5. Review disk_evidence/disk_layout.txt, fdisk_output.txt and filesystem_usage.txt.
6. Review disk_evidence/raid_calculations.txt.
7. If both SSD and HDD are available, rerun with SSD_DIR and HDD_DIR set to writable mount

directories.

8. Fill the worksheet using the generated evidence and capture screenshots.

Evidence checklist: low-level/partition/logical formatting explanation; boot process; bad-block handling; swap-size/location decision; RAID comparison; actual lsblk/fdisk/swap output; 6 x 4 TB capacity calculations; RAID 1/RAID 5 probability calculations; available SSD/HDD read-speed measurement; RAID 6 alternative; and final recommendation.

Source alignment: Q6 - Disk Management, 4 Marks, K5 - Evaluate, pages 29-30 of the supplied Assignment 3 question paper.